Zhengdong Hong

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Education

Zhejiang University — Hangzhou, China
Ph.D. in Computer Science, State Key Laboratory of CAD&CG at Zhejiang University — *Sep 2021 – Jun 2026*
B.Sc. in Electrical and Electronics Engineering; ***Rank: 1/48*** — *Sep 2017 – Jun 2021*

Electrical Engineering Excellence Program: I spent 6 days a week in hardware design and debugging laboratory, independently learning and building embedded hardware and software systems.

Research Experience

Amazon Frontier AI & Robotics — San Francisco, USA
Research Scientist Intern — *May 2026 – Present*

- Working on **agentic open-world mobile manipulation** on a wheeled humanoid — from robot bring-up to an end-to-end autonomous system, fully onboard.

University of California, San Diego, Prof. Hao Su's Group — UC San Diego, USA
Junior Specialist, Research Scholar — *Aug 2023 – Dec 2024*

- Lead the dexterous manipulation project *Learning Particle-based World Model from Human for Dexterous Manipulation* in Prof. Hao Su's group.
- Published a first-author IROS 2024 paper *EasyHeC++: Fully Automatic Hand-Eye Calibration with Pretrained Image Models*, advised by Prof. Hao Su.
- Responsible for hardware design and maintenance in Prof. Hao Su's Group since November, 2023.

Zhejiang University — Hangzhou, China
3D computer vision (dynamic scene reconstruction) — *Sep 2021 – Aug 2023, Jan 2025 – now*

- Independently published a first-author ECCV 2024 paper *Free-Viewpoint Video of Outdoor Sports Using a Flying Camera* project page, which is focused on human pose estimation and 4D scene reconstruction.

Awards

Championship in 8th Texas Instruments Electronic Design Competition — Texas Instruments (TI)
Ranking: 1/1847 (TI Cup, Highest Award) — *Nov 2020*

- In 2020, I lead my team and championed 8th Texas Instruments (TI) Electronic Design Competition, where I lead the whole process of the design and evaluation of Amplifier Nonlinear Harmonic Distortion Analysis System.

Publications

- **Zhengdong Hong***, Shenrui Wu*, Haozhe Cui*, Boyi Zhao, Yiyang He, Hangxing Zhang, Ran Ji, Zundong Ke, Jun Wang, Guofeng Zhang, Jiayuan Gu, *R3D: Rethinking 3D Policy Learning.* project page
- Jingjing Wang, **Zhengdong Hong**, Chong Bao, Yuke Zhu, Junhan Sun, Guofeng Zhang, *LAMP: Lift Image-Editing as General 3D Priors for Open-world Manipulation.* project page
- **Zhengdong Hong***, Yulin Liu*, Haowen Hou, Bo Ai, Jun Wang, Tongzhou Mu, Yuzhe Qin, Jiayuan Gu, Hao Su, *Learning Particle-based World Model from Human for Dexterous Manipulation*, RSS 2025-Dex (Oral). venue page
- **Zhengdong Hong**, Guofeng Zhang, *Learning Object-Centric Motion Priors from Human for Robotic Dexterous Manipulation*, AAAI 2026 (Oral). project page
- Liangzhi Shi, Yulin Liu, Lingqi Zeng, Bo Ai, **Zhengdong Hong**, Hao Su, *Learning Adaptive Dexterous Grasping from Single Demonstrations*, IROS 2025 (Oral). project page
- **Zhengdong Hong**, *Free-Viewpoint Video of Outdoor Sports Using a Flying Camera*, ECCV 2024. project page
- **Zhengdong Hong***, Kangfu Zheng*, Linghao Chen, Hao Su, *EasyHeC++: Fully Automatic Hand-Eye Calibration with Pretrained Image Models*, IROS 2024 (Oral). project page

Robot Systems

I have **both software and hardware** skills, which enables me to independently build several complete robot systems for Prof. Hao Su's Lab, including algorithm, hardware and system design. see details.

- **Su Lab's first mobile manipulator** equipped with a xArm and an ability hand (onboard power supply).
- **Dexterous manipulation platform** with 2 xArm7 and 2 ability hands for articulated object manipulation.

Hardware Skills

I have plenty of hardware experiences, see details here. Here are some of my hardware skills:

- **Electronics:** electronic design, PCB design, PSpice electronic simulation, power design.
- **Embedded systems:** DSP, PLC, other microcomputers (STM32, MSP430, etc.)
- **Hardware debugging and electronic design for Labs**, see this document for details.